

SIMS

Student Intern Management System

Complete Project Specification · AI Features · Execution Blueprint

Core Stack

React 18 · MUI v6 · Tailwind CSS · React Router v7 · Django REST Framework

AI Layer: Anthropic Claude API · LangChain · Python NLP

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1. System Overview

SIMS (Student Intern Management System) is a full-stack, AI-augmented web application that acts as a lightweight ERP for managing the complete intern lifecycle — from entity setup and onboarding through daily operations, task execution, AI-powered coaching, performance evaluation, and exit. A standalone AI layer runs alongside every core module to automate insights, assist interns, and reduce manual effort for managers and mentors.

1.1 Core Objectives

- Manage the full intern lifecycle across multiple organizations from a single platform
- Enforce granular role-based access — every user sees and does only what is permitted
- Automate monitoring: attendance tracking, task deadlines, escalations, and performance metrics
- Provide AI-powered features: performance analysis, resume tools, mock interviews, chatbot assistant, and smart recommendations
- Generate certificates, reports, and analytics without manual effort

1.2 Three Core Verticals

Vertical	Modules	Purpose
Identity & Access	Auth, Roles, Permissions, Entities	Who can log in, what they see and do, and which organization they belong to
Operations	Onboarding, Attendance, Tasks, Payments, Assets, Documents, Leave, Learning	Day-to-day management of interns from joining to exit
Intelligence	Feedback, Metrics, Reports, Certificates, AI Features, Notifications	Measuring outcomes, AI coaching, generating insights, and communicating results

2. Role Hierarchy & Responsibilities

The system is governed by a strict six-level role hierarchy. Higher roles inherit broader access; lower roles are restricted to their immediate context.

Role	Scope	Key Capabilities	Data Visibility
Super Admin	All entities system-wide	Create entities, assign Managers, activate global modules	Unrestricted across all entities
Manager	Single entity	Full entity control: users, modules, payments, reports, configuration	All data within the entity
Lead / SME	Domain within entity	Create projects, assign tasks, performance evaluation, domain analytics	Domain-level — all interns under domain
Mentor	Assigned interns only	Assign/review tasks, approve leave, submit feedback, view attendance	Only interns assigned to them
Staff (Intern view)	Individual intern record	Full profile, documents, tasks, payments for managed intern	Single intern full record
Intern (Self)	Own account only	Check in/out, tasks, leave, documents, payments, AI tools	Own profile and data only

2.1 Permission Framework

Every action is gated by the permission engine. Four top-level dashboard access flags are checked at route level: `hasInternAccess`, `hasAssetAccess`, `hasAttendanceAccess`, `hasPayrollAccess`. Within each dashboard, fine-grained permissions control individual actions (create, edit, delete, approve, export). The permission check is `GET /Sims/user-permissions/` — called on every protected route mount. Data visibility is enforced server-side by scoping all queries to the requesting user's entity, role, and assigned interns/domains.

3. System Architecture

3.1 Frontend Architecture

React 18 SPA organized as independent dashboard shells — each major functional area has its own sidebar and renders sub-views conditionally based on active view state, eliminating full page reloads. MUI v6 provides the component library; Tailwind CSS handles layout and responsive utilities.

Dashboard Shell	Route Prefix	Who Uses It
Admin Dashboard	/admin/*	Super Admin and Manager — entity management and overview
Intern Dashboard (staff)	/intern/*	Lead, Mentor, Manager — viewing/managing intern records
Task Dashboard	/task/*	Lead, Mentor, Manager — project and task management
Attendance Dashboard	/attendance/*	All staff with attendance access
Asset Dashboard	/asset/*	All staff with asset access
Payroll Dashboard	/payroll/*	All staff with payroll access
Intern Self-Service	/intern-user/*	Intern — personal dashboard, tasks, AI tools

3.2 Backend Architecture

The backend is a Django REST Framework application. All API endpoints are prefixed with /Sims/ for the main SIMS app and /api/ for the standalone Forms app. Authentication uses DRF Token auth. The backend handles all business logic, permission enforcement, data aggregation, certificate generation, and file storage.

3.3 AI Architecture

The AI Layer is a separate Django app (or microservice) that integrates with the Anthropic Claude API (claude-sonnet-4-5 model). AI endpoints are prefixed with /ai/ and are secured by the same token auth as the rest of the system. Every AI call includes the user's context (role, domain, metrics, recent history) as the system prompt so the model gives personalized, data-grounded responses.

Layer	Responsibility
Django REST API (/Sims/ + /api/)	Authentication, CRUD operations, business logic, data aggregation, permission enforcement, file storage, certificate generation

Layer	Responsibility
AI Service (/ai/)	LLM calls to Claude API, resume parsing, interview simulation, performance analysis, smart recommendations, chatbot responses, skill gap analysis
Frontend (React)	UI rendering, state management, API calls via centralized Axios instance, real-time updates via polling

3.4 API Layer

The frontend communicates exclusively via a centralized Axios instance (src/services/api.js) which:

- Attaches the auth token to every outgoing request header
- Intercepts 401 responses globally and redirects to /loginpage, clearing stored tokens
- Has a 30-second timeout to fail fast on unresponsive endpoints
- Is organized into named API modules: authAPI, usersAPI, tasksAPI, attendanceAPI, assetsAPI, feesAPI, documentsAPI, certificatesAPI, feedbackAPI, notificationsAPI, formsAPI, dashboardAPI, and more
- AI endpoints use same token auth — no separate AI login required
- AI responses are streamed where applicable for real-time UX (chatbot, mock interview, resume builder)

3.5 State Management

State is managed entirely within React components using useState and useEffect hooks. There is no global state library (no Redux, no Zustand). Shared state is passed via props or React Context. Each dashboard shell manages its own active view, data cache, and loading state locally.

4. Entity & Organizational Structure

Supports a hierarchical organizational model — a single SIMS installation serves multiple independent companies or academies, each with their own staff, interns, configuration, and data silo.

Level	Created By	Description
Entity (Company / Academy)	Super Admin	Top-level container. All data belongs to an entity. Each entity has one Manager assigned.
Branch	Manager	Physical or logical sub-division (e.g., Chennai Office, Remote Branch).
Department	Manager	Functional grouping within a branch (e.g., Engineering, Design, Data Science).
Domain	Manager / Lead	Specialization within a department (Full Stack, ML, DevOps, UI/UX, Cloud, Security, Data Science).
Team	Manager / Lead	Named group of interns assigned to a Mentor for project work.

4.1 Hierarchy Levels

Level	Created By	Description
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4.2 Setup Flow

1. Super Admin creates an Entity and assigns a Manager.
2. Manager creates Branches, Departments, and Domains.
3. Manager creates Staff accounts (Lead, Mentor) assigned to departments.
4. Manager creates Teams and assigns Mentors as team leads.
5. Interns are onboarded and assigned to a Domain and Team.

4.3 Entity Configuration

The Manager can configure the following at entity level via the Settings module:

- Working hours and shift timings (shift_timing field set at intern onboarding)
- Leave policy: quota, carry-forward rules, approval chain (Mentor → Lead → Manager)
- Payment cycles and fee structures (weekly, bi-weekly, monthly)
- Feature module activation / deactivation per entity (Learning Phase, Stipend, task self-creation, AI features)
- SLA default rules: default time window for task completion warnings and escalation thresholds
- File upload restrictions: allowed types (PDF, JPG, PNG) and maximum file sizes
- Notification templates: customize content of system-generated emails and in-app messages

5. Authentication & Session Management

5.1 Login Flow

6. User visits /loginpage and submits credentials to POST /Sims/login/ — returns Token.
7. Frontend stores token in localStorage. Username and role also stored.
8. GET /Sims/user-permissions/ is called → user routed to their permitted dashboard.

5.2 Route Protection

All routes are wrapped in ProtectedRoute (checks token presence) or ProtectedDashboardRoute (additionally calls /Sims/user-permissions/ and redirects unauthorized users). 401 responses auto-clear localStorage and redirect to /loginpage.

5.3 Password Reset Flow

9. User visits /Recovery — enters email — POST /Sims/password-reset/request/ sends OTP.
10. User enters OTP — POST /Sims/password-reset/verify/.
11. User sets new password — POST /Sims/password-reset/update/.

5.4 Logout

Logout calls POST /Sims/logout/, then clears localStorage (token, username, role) and redirects to /loginpage. The global 401 Axios interceptor also auto-logouts the user if any API call returns unauthorized, ensuring stale sessions are cleared immediately.

5.5 Security Hardening

- MFA (TOTP/OTP) for Manager, Lead, and Super Admin roles
- Session management: active session list, force logout of specific sessions
- Password complexity enforcement (min length, mixed characters) and expiry reminders
- Role delegation with time expiry — delegated permissions auto-expire; all delegate actions are attribution-logged
- Immutable audit log (append-only) for all write operations

6. User Management

6.1 Staff Account Creation

Manager creates staff accounts (Lead, Mentor, Admin) via `/admin/register`. Fields: name, email, employee ID, department, role, shift timing. `POST /Sims/register/` creates the Django auth user and associated profile record.

6.2 Staff Management

The Staff List page (`/admin/user`) provides the Manager with a full table of all staff members with:

- View full staff profile (personal details, college details, documents)
- Edit staff details via `PATCH /Sims/user/update/{empId}/`
- Deactivate / soft-delete staff via the `deleted-users` endpoint
- Filter by department, role, and status
- View staff grouped by department via `GET /Sims/staffs-by-department/`

6.3 Intern Account Creation — via Onboarding Form

Interns are created through a dedicated multi-step onboarding form at `/InternOnboarding`. The form collects the following data grouped into logical sections:

6.3a Intern Onboarding Form — Fields

Section	Fields Collected
Personal Details	Employee ID, Full Name, Email, Phone, Aadhar Number, Gender, Date of Birth, Photo upload
Academic Details	Registration Number, Location, College Name, Degree, College Department, Year of Passing
Internship Details	Start Date, End Date, Shift Timing, Scheme (Free / Paid / Stipend), Department, Domain, Terms agreement

6.3 Onboarding Approval Flow

12. Intern submits form → status: Pending.
13. Manager reviews Onboarding List (`GET /Sims/onboarding/list/`).
14. Manager enables account (`POST /Sims/onboarding/enable/{responselId}/`).
15. Manager sends credentials (`POST /Sims/onboarding/send_credentials/{internId}/`).
16. Status → Active.

6.4 Intern Onboarding Approval

Managers review pending onboarding submissions at the Onboarding List page (/admin/interns or OnboardingList.js). The approval flow:

17. Intern submits onboarding form → status becomes Pending.
18. Manager views the Onboarding List: GET /Sims/onboarding/list/.
19. Manager enables the account: POST /Sims/onboarding/enable/{responseId}/.
20. Manager sends credentials to intern: POST /Sims/onboarding/send_credentials/{internId}/.
21. Intern status moves to Onboarded / Active.

6.5 Profile Updates

Interns can request profile updates which go through an approval workflow:

22. Intern submits update request via POST /Sims/submit-personal-update/.
23. Manager sees pending requests via GET /Sims/profile-update-request/.
24. Manager approves (POST /Sims/approve-profile/{pk}/) or rejects (POST /Sims/reject-profile/{pk}/).

6.6 Bulk Operations

- CSV bulk import of interns with field mapping, validation, duplicate detection, and error log
- Bulk intern status updates (e.g., mark batch as completed at term end)
- Bulk task assignment to a cohort of interns
- Bulk asset issue to a batch of onboarding interns

7. Intern Lifecycle

State	Description & Trigger
yettojoin	Registered but start date not yet reached.
active / inprogress	Actively doing the internship. Default after onboarding approval.
onleave	Approved leave covers today. Reflected in attendance calculations.
completed	Internship duration ended or marked complete by Manager/Lead. Archived from active lists.
discontinued	Dropped or terminated. Removed from active operations. Data retained.

7.1 Lifecycle States

State	Description & Trigger
yettojoin	Registered but start date not yet reached. Appears in the 'Yet to Join' count on the Admin Dashboard.
active / inprogress	Actively doing the internship. Default state after onboarding approval. Intern can check in, access tasks, submit leaves.
onleave	Intern has an approved leave that covers today. Automatically reflected in attendance calculations.
completed	Internship duration ended or marked as completed by Manager/Lead. Intern is archived from active lists.
discontinued	Intern dropped or terminated before completion. Removed from active operations. Data retained.

7.2 Promotion Pathway

Free → Paid → Stipend → Employee → Mentor / Lead

Each promotion is recorded with date, performance basis, and the approving Manager or Lead. Promotion history is visible on the intern profile.

7.3 Intern Status Counts on Admin Dashboard

The Admin Dashboard (GET /Sims/admin/dashboard-summary/) aggregates intern counts in real time:

- Total Interns — all non-deleted intern records
- Active — status is inprogress or active
- Completed — status is completed

- Yet to Join — status is yettojoin
- On Leave — status is onleave
- Discontinued — status is discontinued

7.4 Activity Timeline

Every intern has a chronological event feed showing: check-ins, task status changes, leave applications, feedback received, document uploads, payments, promotions, and AI tool usage. Accessible to Mentor and above.

7.5 Archive & Recovery

Completed/discontinued interns are soft-deleted — all records (tasks, attendance, payments, feedback, documents) are retained. Authorized roles can restore from the Recovery page. Permanent purge is a separate Manager/Super Admin action.

8. Attendance & Time Tracking

8.1 Check-in / Check-out (Intern View)

From the intern dashboard (Dash.js), the intern has a real-time attendance widget:

25. Intern clicks Check-in → POST /Sims/attendance/live-on/ → timestamp recorded.
26. Intern can start a break → POST /Sims/attendance/break-start/.
27. Intern ends break → POST /Sims/attendance/break-end/ — break time excluded from effective hours.
28. Intern clicks Check-out → POST /Sims/attendance/live-off/ → system calculates total login hours.

All timestamps are stored in UTC and displayed in the user's local timezone. Total login hours per day are computed by the backend.

8.2 Attendance Dashboard (Staff Views)

View	Description
Dashboard	Overview: total present today, attendance %, on-leave count. Date-filterable.
Daily Attendance	Table of all interns: check-in/out times, total hours, status (Present/Absent/On Leave).
Attendance Log	Date-range filtered log. GET /Sims/attendancedaterange/ and /Sims/simpleattendancedata/.
Leave List	All leave requests with status filter. Staff can approve or reject.
Attendance Claims	Intern correction requests. Staff approves or rejects with reason.

8.3 Leave Management

Intern: Apply for Leave

29. Select leave type, start date, end date, reason → POST /Sims/attendances/leave_request/.
30. Leave appears in history: GET /Sims/attendances/leave_history/.

Staff: Approve/Reject Leave

31. View pending: GET /Sims/attendances/leave_approval/.
32. Approve or reject: PATCH /Sims/attendances/leave_approval/{leaveId}/ with status and comment.
33. Leave days deducted from attendance % calculation.

8.4 Attendance Claims (Correction Requests)

34. Intern submits a claim for a date where attendance was not recorded: POST /Sims/attendance-claims/.
35. Staff reviews: GET /Sims/attendance-claims/pending-approval/.
36. Staff approves (POST .../approve/) or rejects with reason (POST .../reject/).

8.5 Intern Hours Calculator

A dedicated Hours Calculator page (InternHoursCalculator.js) at /intern-user/hours-calculator allows interns and staff to compute total effective working hours over any date range, broken down by day, with leave days clearly separated from worked hours.

8.6 Attendance Analysis

GET /Sims/attendanceanalysis/ provides aggregated attendance statistics used by dashboards and performance metric calculations. GET /Sims/attendancedaterange/ supports date-range filtered views. These endpoints power the attendance charts on the Attendance Dashboard and the Admin Dashboard summary.

8.7 Shift Configuration

- Entity-level shift definitions (morning, evening, standard) configured by Manager in Settings
- Each intern assigned a shift_timing at onboarding
- Late-mark and half-day rules computed based on assigned shift and check-in timestamp
- Flexible shift support for entities with non-standard working hours

9. Task & Project Management

9.1 Task Dashboard Views

View	Purpose
Tasks Dashboard	Overview: counts by status, monthly trend chart, distribution donut, due-today list.
Task List	Full filterable table. Filters: project, status, priority, assignee, date range. Bulk actions.
Projects	Project cards with status, team, task count. Full CRUD.
Project Status	Per-project detail: task breakdown, team members, progress bar, milestones.
Teams	Team cards with interns and Mentor. Create/edit teams, assign/remove interns.
Individual Task	Deep-dive: description, subtasks, status history, comments, file attachments.
Departments	Department and domain hierarchy visualization.
User List	All users accessible from task context.

9.2 Task Data Model

Field	Description
Title / Description	Task name and detailed requirements.
Status	To Do → In Progress → Completed.
Priority	Low / Medium / High — color-coded badges.
Task Type	Learning / Project / Custom.
Assigned To	Responsible intern or staff member.
Project	Parent project FK.
Due Date / Committed Date	Deadline for SLA calculation and escalation sorting.
Progress	0–100%. Set to 100% when Completed.
blocked_by	FK to prerequisite task — dependency guard on status transition.
sla_hours	SLA time window in hours.
Subtasks	Child tasks with own status.

9.3 Task Creation Flow

37. Lead or Mentor opens 'Create Task' from the Task List view (CreateTaskDialog.js).
38. Fills in title, description, project, assigned intern/staff, priority, due date, and task type.
39. Optionally adds subtasks inline.
40. Submits → POST /Sims/tasks/.
41. Task appears in the Task List and on the assigned intern's dashboard.
42. Intern receives an in-app notification of the new assignment.

Bulk project task creation: The TaskManagerCreation component allows creating multiple tasks for a project at once via a spreadsheet-style grid interface.

9.4 Task Status Flow

To Do → In Progress → Completed (Pending Review) → Verified

Intern marks task done → enters completion review queue → Mentor/Lead verifies via GET /Sims/completion-review/. PATCH /Sims/tasks/{pk}/update-status-auto-next/ moves to next status.

9.5 Task Dependency & SLA

- Blocked-by linking: task cannot move to In Progress until all prerequisite tasks are Completed
- SLA deadline fields with visual yellow warning (approaching) and red breach indicator
- Auto-escalation engine: overdue task notifies intern → if unresolved, escalates to Mentor → Lead → Manager, each after a configurable delay
- Inactivity detection: intern with no check-in or task activity within configurable timeout triggers alert chain

9.6 Project Management

Projects group related tasks and team members. Project operations:

- Create project: POST /Sims/projects/ — set name, description, domain, start/end dates
- Assign team to project: POST /Sims/projects/{pk}/assign-team/
- Assign team lead: POST /Sims/projects/{pk}/assign-team-lead/
- Project Dashboard: GET /Sims/projects/dashboard/ — aggregated stats per project
- Team Lead view: GET /Sims/team-lead/projects/ — projects visible to the current team lead

9.7 Team Management

Teams are managed from the Teams view within the Task Dashboard:

- Create team: POST /Sims/teams/ — assign a Mentor as team lead
- Assign interns to team: POST /Sims/teams/{teamId}/assign-interns/
- Remove intern from team: POST /Sims/teams/{teamId}/remove-intern/

- Available interns (not yet in a team): GET /Sims/teams/available-interns/
- Team interns page (TeamInternsPage.js): lists all interns in a specific team with status

9.8 Intern Task View (Self)

From the intern's own dashboard (Dash.js), the intern sees:

- Task cards for all their assigned tasks grouped by status (To Do, In Progress, Completed)
- Due-today tasks highlighted at the top via GET /Sims/tasks/due-today/
- Ability to update task status and progress on each card
- Subtask management within each task card
- Weekly performance stats via GET /Sims/tasks/weekly-performance/
- Monthly task count trend via GET /Sims/tasks/monthly-count/

9.9 Project & Team Management (Combined)

- Project CRUD: POST /Sims/projects/ — assign team (POST .../assign-team/) and team lead (POST .../assign-team-lead/)
- Team CRUD: POST /Sims/teams/ — assign interns (POST .../assign-interns/), remove intern (POST .../remove-intern/)
- Bulk project task creation via spreadsheet-style grid (TaskManagerCreation component)
- Gantt chart view for project task planning with dependency visualization

10. Learning / Training Phase

A dedicated pre-project phase isolated from active project work. Learning tasks do not count towards project productivity KPIs but are tracked separately for intern progress.

10.1 Purpose

- Provide structured onboarding education before assigning project tasks
- Allow Mentors to assign practice exercises and reading material without affecting KPIs
- Give Leads and Managers visibility into how quickly interns progress through the learning curriculum

10.2 Learning Task Flow

43. Mentor assigns learning tasks (Task Type = Learning) to a new intern.
44. Intern accesses dedicated Learning section in their dashboard — separate from project tasks.
45. Intern works through materials and marks each learning task complete.
46. Mentor monitors progress (% completion) on their Intern Management Dashboard.
47. Mentor marks learning phase complete → intern transitions to Active project status.

10.3 Learning Path Assignment

Mentors can assign an ordered sequence of learning tasks as a learning path. Sequential dependencies are enforced — task N+1 unlocks only after task N is completed. Interns cannot skip ahead.

10.4 AI-Enhanced Learning (see Section 15.3)

AI features overlay onto the learning phase: skill gap analysis, personalized learning path recommendations, auto-generated practice quizzes, and AI-powered concept explanations triggered from within learning tasks.

11. Payment & Stipend Management

11.1 Payment Types

Type	Description
Free	No stipend. Intern does not receive payment. Fee structure may apply (intern pays).
Paid	Fixed payment set by Manager at onboarding or after promotion.
Stipend	Performance-based. Amount may vary based on attendance and task completion metrics.

11.2 Payroll Dashboard Views

- Dashboard: Summary cards (total, pending, completed, overdue). Payment trend chart.
- Payment Management: Full table of all intern payment records. Filter by status, payment method. Search by intern name.
- My Profile: Staff profile within the payroll context.

11.3 Payment Record Operations

- View payment details: GET /Sims/fees/{empld}/ — per-intern payment record
- Create payment record: POST /Sims/fees/
- Update payment record: PATCH /Sims/fees/{pk}/ — update amount, mode, status
- Fee structures: GET/POST /Sims/fee-structure/ — configurable payment tiers

11.4 Payment Approval Workflow

Cash payments require additional approval:

48. Lead or Manager marks a payment as Cash mode.
49. Payment enters a Pending Approval state.
50. Manager reviews and approves or rejects the cash payment record.
51. Approved payments are marked as Paid. All changes are logged with old/new values.

11.5 Intern Payment View (Self)

Interns can view their own payment status from the Payment Status page (PaymentStatusPage.js) at /intern-user/payment-status. They can see: payment type, amount, status (Paid / Pending / Overdue), and payment mode. Interns cannot edit payment records.

11.6 Payment Cycle Configuration

- Payment cycle configuration: weekly, bi-weekly, monthly per entity
- Payment history versioning: all edits logged with old/new values and timestamps
- Fee structure configuration: tiered structures per entity for different intern schemes

11.3 Intern Payment View (Self)

Read-only view of payment type, amount, status (Paid/Pending/Overdue), and payment mode. Interns cannot edit payment records.

12. Asset Management

12.1 Asset Dashboard

The Asset Dashboard (/asset/*) gives staff full visibility over all physical assets. Views include: Dashboard (counts, trend chart, type distribution), Asset Management (full CRUD table), Intern Status (which interns hold which assets), and Asset Reports (history, issue log).

12.2 Asset Data Model

- Asset Code — unique identifier (used for lookup: GET /Sims/asset-lookup/{code}/)
- Asset Type — Laptop, ID Card, Accessories, etc.
- Serial Number — physical serial number
- Condition — Good, Damaged, Lost
- Status — Available, Assigned, Damaged, Lost
- Assigned To — intern or staff emp_id
- Issue Date and Expected Return Date

12.3 Asset Lifecycle Flow

52. Manager creates asset record (asset code, type, serial number, condition): POST /Sims/assert-stock/.
53. Asset assigned to intern with issue date and expected return date: PATCH /Sims/assert-stock/{pk}/.
54. If damage/loss: POST /Sims/assert-issue/ — issue report created.
55. Replacement asset lookup: GET /Sims/assert-issue/{pk}/replacement-assets/.
56. New asset assigned as replacement: POST .../assign-new-asset/.
57. On return: condition verified by Manager before record closes.
58. Full audit history: GET /Sims/asserthistory/ and /Sims/assertuserhistory/.

12.4 Asset Dashboard Views

View	Description
Dashboard	Total/available/assigned/damaged/lost counts. Monthly trend chart. Type distribution.
Asset Management	Full table: filter by type, status. Create/edit/delete. Assign and return.
Intern Status	Which interns have which assets currently assigned.
Asset Reports	Assignment history, issue log, expected returns.

12.5 Intern Asset View (Self)

Interns can view their own assigned assets and report issues via the Asset Report page (/intern-user/asset-report). They can submit issue reports (POST /Sims/assert-issue/) but cannot modify asset records directly.

13. Document Management

13.1 Document Types

- Offer Letter, NDA (Non-Disclosure Agreement), Resume/CV, ID Proof, Educational Certificates, Other entity-required documents

13.2 Document Upload Flow (Intern)

59. Intern logs in and navigates to Document View (/intern-user/documents).
60. Clicks upload for a required document type.
61. Selects file (PDF, JPG, PNG) and uploads: POST /Sims/documents/.
62. Document status becomes 'Pending Review'.
63. Intern sees uploaded vs pending status for each document type.

13.3 Document Approval Flow (Staff)

64. Staff accesses the Intern Approval Dashboard (/intern/approval-dashboard) or the Approve Dashboard (ApproveDashboard.js).
65. GET /Sims/intern-approval-dashboard/ returns all interns with pending document counts.
66. Staff clicks on an intern to view uploaded documents.
67. For each document: Approve (POST /Sims/approve-document/{pk}/) or Reject with comments (POST /Sims/reject-document/{pk}/).
68. Rejected documents must be re-uploaded by the intern.
69. Approved documents are locked and versioned — re-uploads create a new version.

13.4 Document Storage

Files are stored in entity-scoped folders: entity/{entityID}/interns/{internID}/documents/. Documents can be downloaded via GET /Sims/documents/{pk}/download/ (returns blob). File type and size validation is enforced on upload (allowed types: PDF, JPG, PNG).

13.3 AI-Enhanced Document Features (see Section 15.4 & 15.5)

The AI layer provides automatic resume parsing and evaluation, resume builder assistance, and document completeness checking integrated directly into the document upload flow.

14. Feedback & Performance Evaluation

14.1 Feedback Types

Type	Description
Mentor Feedback	Periodic qualitative feedback + numeric scores on performance, attitude, and progress.
Lead Evaluation	Overall evaluation determining final outcome (Selected / Not Selected).
Public Feedback Form	Shareable link form sent to external stakeholders about an intern or project.
Student-to-Staff Feedback	Interns submit feedback about their Mentor/Lead (/intern-user/staff-feedback).

14.2 Feedback Submission Flow

70. Mentor opens Performance Feedback page (/intern/feedbacks or PerformanceFeedbackPage.js).
71. Selects intern and fills in the feedback form: qualitative notes, scores, and recommendation.
72. Submits: POST /Sims/feedback/.
73. Feedback is visible in PerformanceFeedbackList.js to the Lead and Manager.
74. Interns can view their own feedback outcomes (configurable visibility).

14.3 Auto-Calculated Performance Metrics

Metric	Formula	Source
Attendance %	$(\text{Present Days} / \text{Working Days}) \times 100$	Attendance records
Task Completion Rate	$\text{Completed Tasks} / \text{Total Assigned}$	Task records
Effective Hours Ratio	$\text{Task Timer Hours} / \text{Total Login Hours}$	Attendance + task timers
Quality Rating	Avg of all Mentor + Lead feedback scores	Feedback records
Learning Completion %	$\text{Completed Learning Tasks} / \text{Total Assigned}$	Learning task records

14.4 Performance Page (Intern View)

The intern can view their own performance dashboard from `/intern-user/performance-feedback`. This page (`PerformancePage.js`) shows:

- Monthly attendance trend (line chart)
- Weekly task completion trend (bar chart)
- Current performance metrics (doughnut / circular progress bars)
- Recent feedback entries from Mentors
- Final evaluation outcome (Selected / Not Selected) when published by Lead

These metrics are surfaced via `GET /Sims/performance/`, `GET /Sims/api/intern/monthly-performance/`, and `GET /Sims/tasks/weekly-performance/`.

14.5 Forms & Surveys Module

A standalone forms builder (`/intern/forms`) allows staff to create custom feedback forms and surveys:

- Question types supported: text, textarea, radio, checkbox, dropdown, email, phone, rating (1–5 stars), yes/no
- Forms can be organized into sections with section titles
- A shareable public link (`/feedback/{formId}`) can be sent to external respondents
- Form responses are collected and visualized in `FormResponses.js` and `FormAnalytics.js`
- Banners can be uploaded per form (`POST /api/forms/{formId}/upload_banner/`)
- Consult flag: staff can mark a response for team discussion (`PATCH /api/responses/{responseId}/consult/`)
- Comments can be added to individual responses (`POST /api/responses/{responseId}/comments/`)

14.6 Forms Builder

- Question types: text, textarea, radio, checkbox, dropdown, email, phone, rating (1–5 stars), yes/no
- Forms organized into sections with section titles
- Shareable public link (`/feedback/{formId}`) for external respondents
- Form response analytics and consult flag (mark response for team discussion)
- Comments on individual responses: `GET/POST /api/responses/{responseId}/comments/`

15. AI Features — Complete Specification

SIMS integrates a dedicated AI layer backed by the Anthropic Claude API (claude-sonnet-4-5 model). All AI features communicate via a secured `/ai/` endpoint namespace using the same token authentication as the rest of the system. The AI layer is context-aware — every request includes the intern's profile, domain, metrics, and relevant history so the model can give personalized, data-grounded responses.

AI Architecture Principles

- Every AI call includes the user's context (role, domain, metrics, recent history) as system context
- Responses are streamed where real-time output improves UX (chatbot, interview feedback, resume suggestions)
- AI-generated content is always presented as a suggestion — the user can accept, edit, or discard
- All AI interactions are logged for audit and to allow Mentors/Managers to review AI recommendations
- AI features are role-scoped: interns can access coaching tools; managers access analytics and insights
- Sensitive data (salary, personal IDs) is never sent to the AI layer — only performance and task data

15.1 SIMS AI Chatbot Assistant

The AI chatbot is the central conversational interface within SIMS. It exists on the intern dashboard (already partially built with rule-based logic), and is upgraded to a full LLM-powered assistant backed by the Claude API. It supports voice input (Web Speech API, already implemented) and voice output (Speech Synthesis, already implemented).

□ AI Chatbot Assistant

Who uses it: Intern (primary) — also available in limited form to Mentor and Manager

What it does: Answers questions about attendance, tasks, leaves, performance, payments, and general SIMS navigation. Provides guidance, encouragement, and contextual help. Can navigate the app on the user's behalf.

How it works: User message + context (role, current metrics, recent tasks, leave balance, attendance %) is sent to Claude API. Response is streamed back and displayed in the chat window. Voice input via browser SpeechRecognition API; voice output via SpeechSynthesis. Chat history is maintained per session.

API endpoint(s): POST `/ai/chat/` | GET `/ai/chat/history/`

Chatbot Capabilities

- Answer natural language questions: 'How many leaves do I have left?', 'What tasks are due today?', 'What is my attendance percentage this month?'
- Navigate the app: 'Take me to my tasks', 'Show my payment status', 'Open leave management'

- Provide guidance: 'How do I apply for leave?', 'What does SLA mean?', 'Explain my performance rating'
- Motivation and coaching: detect low performance context and provide encouragement and actionable tips
- Escalation awareness: if intern asks about an escalation alert, explain what it means and what to do
- Voice commands: full voice-in / voice-out interaction with push-to-talk mic button
- Quick action chips: pre-built suggestion buttons shown after each response for common follow-ups
- Multi-turn conversation with session history — remembers context within the session

Staff-Side Chatbot

- Manager/Mentor can ask: 'Which interns have low attendance this week?', 'Summarize feedback for [intern]'
- Get quick summaries of dashboard data via natural language instead of navigating multiple screens
- Ask for report generation: 'Export attendance report for October'

15.2 AI Performance Analysis & Insights

Replaces and extends the static performance metrics page with AI-generated narrative analysis, trend detection, and personalized recommendations.

□ AI Performance Analysis

Who uses it: Intern (own analysis) + Mentor and Lead (all assigned interns) + Manager (entity overview)

What it does: Generates a natural language performance report for each intern, highlighting strengths, areas of concern, attendance patterns, task efficiency trends, and personalized improvement recommendations.

How it works: Backend aggregates the intern's metrics (attendance %, task completion rate, effective hours ratio, quality rating, feedback history, learning progress) and sends them to Claude with a structured analysis prompt. Claude returns a narrative report with: Summary, Strengths, Areas for Improvement, Trend Analysis, and Action Plan. Reports are cached for 24 hours.

API endpoint(s): POST /ai/performance-analysis/{empId}/ | GET /ai/performance-report/{empId}/

Analysis Output Sections

Section	Content
Executive Summary	2–3 sentence overview of the intern's overall performance standing.
Strengths	Specific, data-backed strengths with examples from task and attendance records.
Areas for Improvement	Specific gaps with concrete suggestions — not generic advice.

Section	Content
Attendance Analysis	Pattern detection: late check-ins, frequent short sessions, leave clustering.
Task Efficiency Trend	Week-over-week task completion trend, quality trend, and speed analysis.
Skill Development	Progress through domain-specific learning tasks and knowledge gaps identified.
Risk Flags	AI flags: at-risk intern (low metrics + declining trend), overloaded intern (too many tasks), disengaged intern (low hours despite availability).
Recommended Actions	Concrete next steps for the intern AND for the Mentor (e.g., 'Schedule a 1:1 this week', 'Reassign 2 tasks to reduce load').

AI Performance Scoring

AI computes a composite AI Performance Score (0–100) for each intern by weighting the four auto-calculated metrics with domain-specific calibration. This score is displayed as the primary performance indicator on dashboards.

Component	Weight	Notes
Attendance %	25%	Below 75% triggers risk flag regardless of other scores
Task Completion Rate	30%	Adjusted for task difficulty if difficulty tags are used
Effective Hours Ratio	20%	Low ratio with high login hours = disengagement signal
Quality Rating (Feedback)	25%	Latest 3 feedback entries weighted more than older ones

Comparative Analytics (Manager View)

- Domain-level performance comparison: average scores across all interns in each domain
- Cohort analysis: this batch vs previous batches on same metrics
- Top performer and at-risk intern identification with AI-generated reasoning
- Mentor effectiveness: correlation between Mentor's feedback quality and intern performance outcomes

15.3 AI Learning Path & Skill Gap Analysis

□ AI Learning Path Generator & Skill Gap Analysis

Who uses it: Intern (receives plan) + Mentor (reviews and approves plan)

What it does: Analyzes the intern's domain, completed learning tasks, quiz scores, and task performance to identify skill gaps and auto-generate a personalized learning path with specific resource recommendations.

How it works: Intern's domain, completed/pending learning tasks, task performance data, and optionally the intern's resume skills are sent to Claude. Claude returns: a skill gap assessment mapped to the domain's required competencies, a prioritized learning path with estimated time per module, and resource type recommendations (read, watch, practice). Mentor can approve, edit, or regenerate the plan.

API endpoint(s): POST /ai/learning-path/generate/{empId}/ | GET /ai/learning-path/{empId}/ | PATCH /ai/learning-path/{empId}/

AI Learning Features

- Skill Gap Report: maps intern's current skills (from resume + completed tasks) against domain competency framework
- Personalized Learning Path: ordered list of learning modules tailored to the intern's gaps and learning pace
- Auto-generated Practice Quizzes: Claude generates MCQ/short-answer quizzes from learning task content on demand
- Concept Explainer: intern can highlight any term in a task description and ask AI to explain it in simple terms
- Learning Pace Detection: AI detects if intern is moving too slowly and alerts Mentor, or moving very fast and suggests advanced topics
- Domain Readiness Score: AI scores how 'project-ready' an intern is based on learning progress and quiz performance

15.4 AI Resume Evaluation

□ AI Resume Evaluator

Who uses it: Intern (self-evaluation and improvement) + Manager/Lead (candidate screening during onboarding)

What it does: Parses and evaluates the intern's uploaded resume against the domain they are applying for or currently working in. Provides a structured score, identifies gaps, and gives specific, actionable improvement suggestions.

How it works: Intern uploads resume (PDF/DOCX). Backend extracts text and sends it to Claude along with the target domain competency framework. Claude returns: ATS compatibility score, section-by-section evaluation (Summary, Experience, Skills, Education, Projects), domain relevance score, missing keywords list, and a rewrite suggestion for weak sections.

API endpoint(s): POST /ai/resume/evaluate/ | GET /ai/resume/evaluation/{empId}/ | POST /ai/resume/evaluate-for-domain/{domain}/

Resume Evaluation Output

Evaluation Component	Details
Overall Score (0–100)	Composite of ATS compatibility, relevance, completeness, and formatting.
ATS Compatibility Score	Keyword density, formatting issues that fail ATS parsers, section headers check.
Domain Relevance Score	How well skills and experience match the target domain (e.g., Full Stack, ML, DevOps).
Section Scores	Individual scores for: Summary/Objective, Skills, Experience/Projects, Education, Certifications.
Missing Keywords	Domain-specific keywords and tools that are absent from the resume.
Formatting Issues	Font consistency, spacing, length (ideal: 1 page for fresher), use of action verbs.
Improvement Suggestions	Specific, per-section rewrites — not generic advice. E.g., 'Rewrite this bullet with a measurable outcome.'
Strengths Identified	What the resume does well — important for intern confidence alongside critique.

Manager Use: Candidate Screening

- During onboarding approval, Manager or Lead can trigger AI evaluation of uploaded resume against the assigned domain
- AI provides a domain fit score and highlights strengths/gaps — helps Mentor plan the learning path immediately
- Multiple resumes can be batch-evaluated for cohort onboarding

15.5 AI Resume Builder

□ AI Resume Builder / Generator

Who uses it: Intern

What it does: Helps interns build a professional, domain-optimized resume from scratch or improve an existing one. Generates tailored resume content based on the intern's profile, tasks completed, skills, and internship achievements recorded in SIMS.

How it works: The AI has access to the intern's SIMS data: tasks completed (with descriptions), attendance record, learning path completed, domain, performance metrics, and any skills the intern manually inputs. Claude generates professional resume content for each section, formatted as ready-to-use text. The intern can refine any section in a chat-like interaction. Final resume can be exported as a PDF.

API endpoint(s): POST /ai/resume/generate/{empId}/ | POST /ai/resume/section/{section}/ | POST /ai/resume/export/{empId}/

Resume Builder Sections & AI Capabilities

Resume Section	AI Capability
Professional Summary	Generates a 3–4 line professional summary based on domain, internship duration, and key achievements from SIMS task data.
Skills Section	Auto-populates technical and soft skills based on: domain, completed learning tasks, task tags, and manual skill input.
Internship Experience	Converts completed SIMS tasks and projects into professional bullet points with action verbs and measurable outcomes (e.g., task completion rate, hours, project names).
Projects Section	Generates project descriptions from SIMS project data: project name, role, technologies used, outcome.
Education	Populated from intern's academic details (college, degree, year) stored in SIMS.
Achievements / Certifications	Populated from SIMS certificates generated (completion, task, attendance).
Keyword Optimization	AI inserts domain-relevant keywords to improve ATS performance without making the resume feel keyword-stuffed.
Tone & Style Adjustment	Intern can specify: fresher / experienced, formal / modern, concise / detailed — AI adjusts writing style accordingly.

Resume Iteration Flow

- 75. Intern opens Resume Builder from their dashboard (/intern-user/resume-builder).
- 76. AI pre-fills all sections using SIMS data. Intern reviews each section.
- 77. Intern can click 'Improve this section' on any section — AI provides alternative versions.
- 78. Intern can type custom instructions: 'Make the summary more confident', 'Add more technical keywords'.
- 79. Intern exports final resume as PDF: POST /ai/resume/export/{empId}/ — server-side PDF generation.
- 80. Resume is saved in SIMS document storage and can be replaced in the Documents section.

15.6 AI Mock Interview

□ AI Mock Interview

Who uses it: Intern

What it does: Provides a fully interactive mock interview experience. The AI acts as an interviewer, asking domain-relevant technical and behavioral questions, evaluating answers in real time, and providing detailed feedback at the end.

How it works: Intern selects interview type and domain. Claude generates a structured interview session (intro → technical questions → behavioral questions → scenario questions → closing). Each intern answer is evaluated by Claude immediately with a score and short feedback. At the end, a comprehensive interview report is generated with overall score, per-question breakdown, confidence assessment, and improvement plan.

API endpoint(s): POST /ai/interview/start/ | POST /ai/interview/answer/{sessionId}/ | GET /ai/interview/report/{sessionId}/ | GET /ai/interview/history/{empId}/

Interview Types

Interview Type	Description
Technical — Domain	Questions specific to the intern's domain: Full Stack (React, Django, REST APIs), ML (Python, scikit-learn, model evaluation), DevOps (CI/CD, Docker, Kubernetes), etc.
HR / Behavioral	Standard behavioral questions: STAR method answers, situational judgment, team scenarios, conflict resolution.
Mixed (Technical + HR)	Combined session simulating a real interview: technical depth first, then behavioral rounds.
Project-Based	Interview questions generated from the intern's actual SIMS projects — 'Tell me about a challenging task in Project X'.
Company Preparation	Intern specifies a target company/role — AI tailors questions to that company's known interview style and tech stack.

Interview Session Flow

- 81. Intern opens Mock Interview from dashboard (/intern-user/mock-interview).
- 82. Selects interview type, domain, difficulty (Beginner / Intermediate / Advanced), and duration (15, 30, 45 minutes).
- 83. AI introduces itself and starts the interview with an opening question.
- 84. Intern types (or speaks via voice input) their answer.
- 85. AI evaluates the answer instantly: score (1–10), brief immediate feedback, and asks the next question.
- 86. Session proceeds for the configured duration or number of questions.
- 87. At end: AI generates a full Interview Report (see below).
- 88. Report saved to intern's AI history and visible to Mentor.

Interview Report

Report Section	Content
Overall Score	Aggregate score across all questions with percentile context (e.g., 'Better than 65% of interns in this domain').
Question-by-Question Breakdown	Each question, the intern's answer, AI score (1–10), and specific feedback on what was good and what was missing.
Strengths Identified	Communication clarity, technical accuracy, problem-solving approach, domain knowledge.

Report Section	Content
Improvement Areas	Specific gaps: 'You didn't mention time complexity in algorithm questions', 'STAR method answers lacked a clear Result step'.
Confidence Indicators	Response time analysis, completeness of answers, use of technical terminology.
Recommended Study Topics	Specific topics to study based on incorrect or incomplete answers.
Action Plan	3–5 specific things to practice before the next interview, with resource suggestions.

Interview History & Progress Tracking

- All past interview sessions stored with scores, dates, and domain
- Score trend chart showing improvement across multiple sessions
- Mentor can view intern's interview history and use it to inform feedback
- Manager can see cohort-level interview readiness distribution

15.7 AI Smart Task Suggestions

□ AI Smart Task Suggestions

Who uses it: Lead and Mentor (creating tasks) + Intern (self-assigning or creating optional tasks)

What it does: When a Lead or Mentor creates a task, AI suggests: task description improvements, appropriate priority and SLA, subtask breakdown, relevant learning resources, and similar past tasks for reference. For interns, AI can suggest additional practice tasks based on skill gaps.

How it works: When task creation dialog opens, AI analyzes the task title and project context, and pre-fills description, suggested priority, estimated hours, and subtasks. Suggestions can be accepted or ignored. AI also checks for dependency conflicts with existing tasks.

API endpoint(s): [POST /ai/tasks/suggest/](#) | [POST /ai/tasks/decompose/](#) | [GET /ai/tasks/similar/{taskId}/](#)

Task AI Capabilities

- Description Generator: enter task title → AI generates a detailed professional task description
- Task Decomposer: enter high-level task → AI breaks it into 3–7 specific, measurable subtasks
- Priority Suggester: AI analyzes task type, project deadline, and intern workload to suggest priority level
- SLA Estimator: based on task type and domain, AI suggests a realistic SLA/due date
- Similar Task Finder: find past tasks similar to the new one for reference and time estimation
- Workload Balancer: when assigning, AI checks intern's current task load and flags overallocation
- Auto-tagging: AI suggests domain and skill tags for the task enabling skill-based reporting

15.8 AI Attendance Anomaly Detection

□ AI Attendance Anomaly Detection

Who uses it: Mentor and Manager (receives alerts) + Intern (receives personalized insights)

What it does: Automatically detects unusual attendance patterns and raises intelligent alerts — beyond simple threshold checks. Identifies patterns like chronic late check-ins, gradual disengagement, suspicious attendance clustering before deadlines, and predicts at-risk dropout.

How it works: The attendance analysis job runs daily. An ML/LLM analysis is applied to each intern's 30-day attendance history. Claude identifies anomalies, classifies their likely cause (sick pattern, Monday absenteeism, pre-deadline cramming, disengagement), and generates a natural language summary for the Mentor. Severe patterns trigger automated escalation.

API endpoint(s): GET /ai/attendance/anomalies/ | POST /ai/attendance/analyze/{empId}/ | GET /ai/attendance/risk-report/

Anomaly Types Detected

- Chronic Late Arrival: check-in consistently 30+ minutes late — AI differentiates from shift change vs disengagement
- Attendance Clustering: intern marks attendance only on task-due days — sign of selective engagement
- Gradual Decline: month-over-month attendance % declining — early warning before it becomes a problem
- Session Duration Anomaly: logged in for very short sessions (< 1 hour) repeatedly without approved reasons
- Absenteeism Pattern: recurring absences on same weekday — AI flags for manager awareness
- Dropout Risk Score: composite score combining attendance trend + task completion trend + last activity date

15.9 AI Feedback Writing Assistant

□ AI Feedback Writing Assistant

Who uses it: Mentor and Lead (writing feedback)

What it does: Helps Mentors write high-quality, specific, data-backed feedback instead of vague comments. AI drafts feedback based on the intern's actual metrics and task history, which the Mentor can review, edit, and submit.

How it works: When Mentor opens the feedback form for an intern, AI pre-analyzes the intern's performance data (attendance, task completion rate, quality scores, specific task outcomes) and generates a draft feedback narrative. The draft is editable. Mentor can also request: 'Make this more constructive', 'Add specific examples', or 'Write a final evaluation summary'.

API endpoint(s): POST /ai/feedback/draft/{empId}/ | POST /ai/feedback/improve/ | POST /ai/feedback/final-evaluation/{empId}/

Feedback AI Capabilities

- Draft Generator: auto-generates a complete feedback entry from metrics and task data — Mentor edits and submits

- Tone Checker: flags feedback that is too vague ('good job'), too harsh (demotivating language), or biased
- Specificity Improver: 'Add a specific example from their work history to support this statement'
- Final Evaluation Summary: AI generates the overall evaluation text (Selected/Not Selected rationale) for Lead review
- Consistency Checker: AI flags if this Mentor's feedback patterns are inconsistent with performance metrics (e.g., positive feedback but low task scores)

15.10 AI Exit Summary & Career Readiness Report

□ AI Exit Summary & Career Readiness Report

Who uses it: Intern (receives report at internship end) + Manager (reviews before certificate generation)

What it does: At the end of the internship, AI generates a comprehensive exit summary that serves both as an internal performance record and as a career readiness report the intern can use for job applications.

How it works: Aggregates the intern's entire SIMS history: all tasks completed, attendance record, learning path, feedback received, interview practice scores, resume quality score, and certifications. Claude generates a structured exit report and a Career Readiness Score (0–100) with domain-specific context.

API endpoint(s): [POST /ai/exit-report/{empld}/](#) | [GET /ai/exit-report/{empld}/](#) | [POST /ai/exit-report/{empld}/export/](#)

Exit Report Contents

- Internship Overview: timeline, domain, projects worked on, team, Mentor
- Performance Summary: all metrics with trend analysis across the full internship duration
- Skills Acquired: AI-curated list of skills demonstrated through task completion and learning path
- Key Achievements: top 3–5 accomplishments derived from task and project data
- Interview Readiness: mock interview progress summary and current readiness level
- Career Readiness Score (0–100): composite of performance metrics, skill development, and interview practice
- Recommended Next Steps: role suggestions, skills to develop, certifications to pursue
- Exportable as PDF: designed for inclusion in intern's job application portfolio

15.11 AI Smart Search

□ AI Smart Search

Who uses it: All roles

What it does: Global search that understands natural language queries — not just keyword matching. Searches across interns, tasks, projects, assets, documents, and reports, with role-scoped results.

How it works: The search query is processed by Claude to identify intent and extract entities. The backend then performs a structured multi-entity search and returns ranked results. For ambiguous queries, AI asks a clarifying question.

API endpoint(s): POST /ai/search/ | GET /search/?q={query}

Search Examples

- 'Show interns with attendance below 80% in the ML domain this month' — structured report query
- 'Find all overdue tasks assigned to Priya' — multi-entity filtered search
- 'Which projects are behind schedule?' — project status semantic query
- 'List interns who haven't uploaded their NDA' — document status query
- 'Show me feedback for interns completing their internship this month' — lifecycle-aware query

15.12 AI Onboarding Readiness Check

□ AI Onboarding Readiness Check

Who uses it: Manager and Lead (reviewing new onboarding submissions)

What it does: When a new intern onboarding submission arrives, AI automatically checks document completeness, flags missing required fields, evaluates the submitted resume for domain fit, and provides a readiness score to help the Manager prioritize and process submissions faster.

How it works: Triggered when new onboarding submission is received. AI checks: all required fields filled, all required documents uploaded, resume uploaded and evaluates domain fit, academic background matches domain requirements. Returns a readiness checklist and domain fit score. Manager can approve with one click for high-readiness interns.

API endpoint(s): POST /ai/onboarding/check/{submissionId}/ | GET /ai/onboarding/readiness-report/{submissionId}/

15.13 AI Natural Language Report Generator

□ AI Report Generator

Who uses it: Manager and Lead

What it does: Generates any report from natural language requests. Manager types what they need and AI queries the data, generates the report, and optionally exports it.

How it works: Manager inputs a report request in plain English. Claude parses the request, identifies required data fields, calls the appropriate API endpoints, aggregates the data, and generates a formatted narrative report with supporting tables and charts. Report can be exported as PDF or Excel.

API endpoint(s): POST /ai/reports/generate/ | GET /ai/reports/history/

Example Report Requests

- 'Give me a performance summary for all interns in the Full Stack domain for Q3'
- 'Show me which interns are at risk of not completing their internship this month'
- 'Compare attendance rates between Morning and Evening shift interns'
- 'List all tasks that breached SLA in the last 30 days with assignees'
- 'Generate a Mentor effectiveness report showing feedback quality vs intern outcomes'

15.14 AI Notification Intelligence

The notification system is upgraded with AI prioritization and intelligent digest:

- Smart Digest: instead of a flood of individual notifications, AI generates a concise daily digest: '3 tasks need your attention, 2 leave requests pending, 1 intern is at attendance risk'
- Priority Classification: AI classifies incoming notifications as Critical, Attention, Informational — surfacing the most important ones first
- Noise Reduction: AI learns from notification dismissal patterns and reduces notification frequency for low-priority event types the user consistently ignores
- Contextual Summaries: for complex notifications (e.g., escalation alert), AI adds a 1-line context: 'This task has been overdue for 3 days and the intern has been inactive since Tuesday'

15.15 AI Feature API Reference

Endpoint	Method	Feature
/ai/chat/	POST	AI Chatbot — send message, receive LLM response
/ai/chat/history/	GET	Chatbot session history for current user
/ai/performance-analysis/{empId}/	POST	Generate AI performance analysis report
/ai/performance-report/{empId}/	GET	Get cached performance report
/ai/learning-path/generate/{empId}/	POST	Generate personalized AI learning path
/ai/learning-path/{empId}/	GET / PATCH	View or update AI learning path
/ai/quiz/generate/{taskId}/	POST	Generate practice quiz from learning task content
/ai/resume/evaluate/	POST	Evaluate uploaded resume (file upload)
/ai/resume/evaluation/{empId}/	GET	Get latest resume evaluation
/ai/resume/evaluate-for-domain/{domain}/	POST	Evaluate resume against specific domain
/ai/resume/generate/{empId}/	POST	Generate resume from SIMS data
/ai/resume/section/{section}/	POST	AI-improve a specific resume section

Endpoint	Method	Feature
/ai/resume/export/{empId}/	POST	Export AI-generated resume as PDF
/ai/interview/start/	POST	Start mock interview session
/ai/interview/answer/{sessionId}/	POST	Submit answer, receive AI evaluation
/ai/interview/report/{sessionId}/	GET	Get full interview report
/ai/interview/history/{empId}/	GET	All past interview sessions for intern
/ai/tasks/suggest/	POST	AI task description and subtask suggestions
/ai/tasks/decompose/	POST	Decompose task into subtasks
/ai/tasks/similar/{taskId}/	GET	Find similar past tasks
/ai/attendance/anomalies/	GET	Entity-level attendance anomaly report
/ai/attendance/analyze/{empId}/	POST	Analyze individual intern attendance pattern
/ai/attendance/risk-report/	GET	Dropout risk scores for all interns
/ai/feedback/draft/{empId}/	POST	AI-draft feedback from performance data
/ai/feedback/improve/	POST	Improve existing feedback text
/ai/feedback/final-evaluation/{empId}/	POST	Generate final evaluation summary
/ai/exit-report/{empId}/	POST / GET	Generate or retrieve exit summary report
/ai/exit-report/{empId}/export/	POST	Export exit report as PDF
/ai/search/	POST	AI smart search across all entities
/ai/onboarding/check/{submissionId}/	POST	AI onboarding readiness check
/ai/reports/generate/	POST	Natural language report generation
/ai/reports/history/	GET	Past generated reports

16. Certificate Generation

All certificates are generated server-side and returned as downloadable files. No manual formatting required.

Certificate Type	Endpoint	Description
Completion Certificate	POST /Sims/generate-completion-certificate/	Generated at end of internship. Confirms successful completion with performance summary.
Offer Letter	POST /Sims/generate-offer-letter/	Formal internship offer letter at onboarding.
Task Certificate	POST /Sims/generate-task-certificate/	Certifies completion of a specific project or task.
Attendance Certificate	POST /Sims/generate-attendance-certificate/	Certifies attendance record and percentage.
Partial Completion Certificate	POST /Sims/generate-partial-certificate/	For interns who completed part of the internship.
AI-Enhanced Completion Certificate	POST /ai/exit-report/{empId}/export/	Includes AI-generated Career Readiness Report alongside traditional certificate content.

16.1 Admin Dashboard Summary Data

The Admin Dashboard calls GET /Sims/admin/dashboard-summary/?date={YYYY-MM-DD} which returns a single aggregated response containing:

- intern_counts: total, active, completed, yet_to_join, on_leave, discontinued
- attendance: pct (percentage present for the selected date)
- payment_summary: completed, pending, overdue, free counts and amounts
- dept_active_counts: array of {dept, count} for the department bar chart

17. Role-Based Dashboards

17.1 Admin Dashboard (/admin/*)

Route	Component	Purpose
/admin/dashboard	AdminDashboardContent	Summary: intern counts by status, attendance %, payment summary, dept distribution chart. Date-filterable.
/admin/user	StaffList	All staff: view, edit, deactivate.
/admin/interns	InternLists	All interns: status filter, CSV export, bulk actions.
/admin/payments	PaymentList	Payment records for all interns.
/admin/departments	DepartmentManagement	Department and domain CRUD.
/admin/entities	EntityManagement	Entity, branch, entity-department CRUD.
/admin/feedback	PerformanceFeedbackPage	All feedback submissions.
/admin/teams	TeamManagement	Team creation and intern assignment.
/admin/register	RegisterPage	Register new staff members.
/admin/profile	AdminProfile	Admin profile management.

17.2 Intern Management Dashboard (/intern/*)

View	Description
dashboard	Summary cards, intern overview, AI risk flags for at-risk interns.
users	Intern table with status, domain, actions. AI performance badge per intern.
documents	Document upload and approval per intern.
feedbacks	Feedback list with AI draft button.
forms	Form builder and response viewer.
approval-dashboard	Pending document approvals.
completionList	Task completion review queue.

17.3 Task Dashboard (/task/*)

View	Purpose
tasks	Overview: counts by status, monthly trend chart, due-today list, AI workload insights.
task-list	Full table with filters, bulk actions, AI task suggestion button.
projects	Project cards. Create/edit. AI project health indicator.
project-status	Per-project detail: task breakdown, team, progress. Gantt view.
teams	Team composition and intern assignment.
departments	Department/domain hierarchy.
individual-task	Single task deep-dive with subtasks, comments, AI explain button.

17.4 Intern Self-Service Dashboard (/intern-user/*)

Sidebar Item	Component	What the Intern Can Do
Dashboard	Dash.js home	Check in/out, due tasks, announcements, performance summary, AI chatbot.
Attendance	AttendanceManagement	Check in/out, breaks, attendance log.
Document View	DocumentView	Upload documents, view approval status. AI resume evaluation.
Payment Status	PaymentStatusPage	Read-only payment view.
Asset Report	AssetReport	View assigned assets, submit issue reports.
Performance	PerformancePage	AI performance analysis, metrics, feedback.
Leave	LeaveManagement	Apply, view history, balance.
Tasks	Tasks	View/update own tasks. AI task explanation.
Hours Calculator	InternHoursCalculator	Calculate hours for any date range.
Staff Feedback	StudentStaffFeedback	Submit feedback about Mentor/Lead.
Teams	TeamsManagement	View team and Mentor.
Mock Interview	MockInterviewPage	AI mock interview sessions. (new page)
Resume Builder	ResumeBuilderPage	AI resume generator and evaluator. (new page)
Learning	LearningPage	Learning path with AI quiz and explainer. (new page)

17.5 Intern Profile (Staff View)

Clicking on any intern in any list opens the full intern profile (InternProfile.js) which shows:

- Personal details, contact information, and photo
- Academic details (college, degree, year of passing)
- Internship details: domain, scheme, start/end dates, shift timing
- Current lifecycle status with change capability
- Documents list with upload/approve/reject actions
- Offer letter and NDA status
- Asset assignment status
- Payment record summary
- Quick links to task and attendance views for this intern

18. Notifications

18.1 Intern Dashboard Home

The dashboard home page shows:

- Real-time check-in/check-out toggle with timer showing current session duration
- Today's task list: due-today tasks with priority badges
- Announcement board: team announcements from Mentors/Leads
- Performance summary: circular progress bars for attendance%, task completion rate, quality rating
- Notification bell with unread count (in-app notifications)
- Monthly attendance trend chart
- Weekly task completion bar chart
- AI chatbot floating button (bottom-right) with voice input/output

18.2 In-App Notifications

- GET /Sims/notifications/ — fetch unread notifications for current user
- PATCH /Sims/notifications/{id}/mark_read/ — mark single as read
- POST /Sims/notifications/mark_all_read/ — mark all as read
- Triggers: task assignment, status change, leave status, payment change, document approval/rejection, attendance claim decision, escalation alerts, AI report ready

18.3 AI Notification Intelligence

- Smart Digest: daily AI-generated summary instead of individual notification flood
- Priority Classification: Critical / Attention / Informational auto-tagging
- Contextual 1-line summaries on escalation and risk alerts
- Noise Reduction: AI reduces low-priority notifications for users who consistently dismiss them

18.4 Email Notifications

- Leave approved/rejected, document approved/rejected, task assigned, SLA warning, claim decision, credentials sent
- Per-user email notification toggle in Settings

18.5 Notification Preferences

Users can configure notification preferences in Settings (/Settings):

- Email notifications toggle (on/off per account)
- Push notifications toggle (on/off per account)
- Specific event subscriptions configurable per notification type
- AI notification digest preference: full list vs AI summary mode

19. Reporting & Analytics

19.1 Role-Based Dashboards Data

Dashboard	Key Metrics & Charts
Admin / Manager	Intern counts by status, attendance %, payment summary, active interns by department, AI risk flags summary
Task Dashboard	Task counts by status, monthly completion trend, due-today list, AI workload distribution
Attendance Dashboard	Daily present/absent, leave summary, attendance %, anomaly alerts
Asset Dashboard	Total/available/assigned/damaged counts, monthly trend, type distribution
Payroll Dashboard	Payment totals by status, method breakdown, trend
Intern Self	AI performance score, attendance %, task rate, quality rating, monthly/weekly trend charts

19.2 Data Export

- CSV/Excel export from all major list views: interns, tasks, attendance, payments, assets, leave
- AI-generated PDF reports exportable from the Reports module
- Resume export as PDF from AI Resume Builder
- Exit Report / Career Readiness Report exportable as PDF

19.3 Activity Logs

All user actions on major entities are logged via GET /Sims/logs/ and GET /Sims/logs/{empId}/. This provides an audit trail of who did what, when. The log model stores: user_id, table_name, action, old_data, new_data, user_name.

- GET /Sims/logs/ — all activity logs (Manager/Admin view)
- GET /Sims/logs/{empId}/ — per-intern activity log
- Logs are append-only (no edit or delete of log records)
- Immutable audit trail: who changed what, when, old value, new value
- AI interaction log: all AI feature usage recorded for Mentor/Manager visibility

19.4 Intern Task Summary

GET /Sims/intern-task-summary/ provides per-intern task completion stats used in reports and the performance dashboard. GET /Sims/intern-count-by-domain/ provides domain-wise intern distribution for organizational analytics.

20. Settings & Configuration

20.1 User Settings (/Settings)

Every user has access to a personal settings page providing:

- Dark Mode toggle — switches the entire application between light and dark theme (ThemeContext)
- Language selection — UI language preference (LanguageContext). Multilingual support is configurable
- Email notifications toggle
- Push notifications toggle
- Profile picture upload (base64 stored, displayed across the app)
- Password change (current password required)
- Account deletion request
- Data reset option

20.1a Personal Settings Details

- Dark mode toggle — persisted to user profile (not just local state)
- Language selection (multilingual UI support)
- Email and push notification toggles with per-event granularity
- Profile picture upload
- Password change (current password required)

20.2 Entity-Level Configuration

The Manager can configure the following at entity level:

- Working hours and shift definitions (morning, evening, standard, custom)
- Leave policy: annual quota, carry-forward rules, leave types, approval chain depth
- Payment cycles: weekly, bi-weekly, monthly
- Feature module activation/deactivation (Learning Phase, Stipend, task self-creation, AI features)
- SLA default rules: default time window and escalation delay thresholds
- File upload restrictions: allowed types (PDF, JPG, PNG), maximum file size
- Notification templates: customize content of system-generated emails and in-app messages
- AI feature configuration: which AI features are enabled per role within this entity

21. Security & Audit

21.1 Authentication Security

- DRF Token authentication: DRF Token — a unique token per user, invalidated on logout
- Automatic token cleanup: 401 response from any API call clears localStorage and redirects to login
- All routes are protected — unauthenticated requests are redirected to /loginpage
- Permission check on every protected dashboard mount via /Sims/user-permissions/
- MFA (TOTP/OTP) for Manager, Lead, and Super Admin
- Session management: active session list and force logout
- Password complexity enforcement (min length, mixed characters) and expiry reminders

21.2 Data Visibility Enforcement

- Server-side scope enforcement: all data-fetching endpoints filter by the requesting user's entity, role, and assigned interns/domains
- Mentors cannot query interns not in their assignment list
- Leads cannot access data from other domains
- Managers see full entity data; Super Admin sees all entities
- AI endpoints enforce the same scoping — intern data is never sent to the AI layer for users who don't have visibility rights

21.3 Audit Logging

- Append-only audit log: user, entity, action, timestamp, old value, new value
- Accessible via GET /Sims/logs/ (Manager) and /Sims/logs/{empld}/ (per-intern)
- AI interaction audit log: all AI requests logged with user, feature, timestamp, and prompt hash
- Role/permission change versioning

21.4 Soft Delete

Major entities (Users, Interns, Tasks, Assets, Documents) support soft delete — records are flagged as deleted (is_deleted) but retained in the database.

- Soft-deleted records do not appear in normal list views
- Authorized roles can restore soft-deleted records from the Recovery page
- Permanent purge is a separate authorized action available only to Manager/Super Admin
- Deleted users list: GET /Sims/deleted-users/
- Archiving is available for completed/dropped interns and closed projects to keep active views clean

21.5 Role Delegation

- Any approval-rights user can delegate permissions to another user for a time window
- Delegation auto-expires at end date
- Delegate actions logged with both delegate identity and original delegator

22. Complete API Reference

All endpoints use Token authentication. Base: <http://localhost:8000> (dev) / <https://sims-3fsg.onrender.com> (prod). AI endpoints use same auth, base: </ai/>.

22.1 Authentication

Endpoint	Method	Description
/Sims/login/	POST	Login — returns Token
/Sims/logout/	POST	Logout — invalidates token
/Sims/register/	POST	Register user
/Sims/register/{empId}/	POST	Register by employee ID
/Sims/password-reset/request/	POST	Request OTP
/Sims/password-reset/verify/	POST	Verify OTP
/Sims/password-reset/update/	POST	Set new password
/Sims/me/	GET	Current user role and permissions
/Sims/user-permissions/	GET	Dashboard access flags

22.2 Users & Profiles

Endpoint	Method	Description
/Sims/users/	GET	List all users
/Sims/interns/	GET	List all interns
/Sims/interns/stats/	GET	Intern count statistics
/Sims/interns/full-list/	GET	Full intern list with all fields
/Sims/all-user-data/	GET	All user profile data
/Sims/user-data/{empId}/	GET / PATCH	Get or update user data
/Sims/user/update/{empId}/	PATCH	Update user record
/Sims/fulldetail/	GET	Current user full detail
/Sims/staffs-details/	GET	All staff with details
/Sims/staffs-by-department/	GET	Staff grouped by department
/Sims/deleted-users/	GET	Soft-deleted users

Endpoint	Method	Description
/Sims/intern-count-by-domain/	GET	Intern count per domain
/Sims/intern-task-summary/	GET	Per-intern task completion summary
/Sims/personal-data/{empld}/	GET / PATCH	Personal profile data
/Sims/submit-personal-update/	POST	Submit profile change request
/Sims/profile-update-request/	GET	Pending profile change requests
/Sims/approve-profile/{pk}/	POST	Approve profile change
/Sims/reject-profile/{pk}/	POST	Reject profile change
/Sims/college-details/{empld}/	GET / PATCH	College/academic details
/Sims/staffs/	GET	All staff list
/Sims/users/reporting-staff/	GET	Staff reporting to current user

22.3 Organization Hierarchy

Endpoint	Method	Description
/Sims/entities/	GET / POST	List or create entities
/Sims/entities/{pk}/	PATCH / DELETE	Update or delete entity
/Sims/branches/	GET / POST	List or create branches
/Sims/branches/{pk}/	PATCH / DELETE	Update or delete branch
/Sims/entity-departments/	GET / POST	List or create entity departments
/Sims/entity-departments/{pk}/	PATCH / DELETE	Update or delete entity department
/Sims/entity-hierarchy/	GET	Full hierarchy tree
/Sims/departments/	GET / POST	List or create departments
/Sims/departments/{pk}/	GET / PATCH / DELETE	CRUD on department
/Sims/domains/	GET / POST	List or create domains

Endpoint	Method	Description
/Sims/domains/{pk}/	GET / PATCH / DELETE	CRUD on domain
/Sims/domains-by-department/	GET	Domains grouped by department

22.4 Attendance

Endpoint	Method	Description
/Sims/attendance/	GET	All attendance records
/Sims/attendance/{empId}/	GET	Attendance for specific intern
/Sims/attendances/user/	GET	Current user's own attendance
/Sims/attendances/leave_balance/	GET	Leave balance for current user
/Sims/attendancedaterange/	GET	Date-range filtered attendance
/Sims/attendanceanalysis/	GET	Aggregated attendance analysis
/Sims/simpleattendancedata/	GET	Simplified data for charts
/Sims/attendance/live-on/	POST	Check in
/Sims/attendance/live-off/	POST	Check out
/Sims/attendance/break-start/	POST	Start break
/Sims/attendance/break-end/	POST	End break
/Sims/attendances/leave_request/	POST	Submit leave request
/Sims/attendances/leave_history/	GET	Own leave history
/Sims/attendances/leave_history/{empId}/	GET	Leave history for specific intern
/Sims/attendances/leave_approval/	GET	Pending leave for approval
/Sims/attendances/leave_approval/{leaveId}/	PATCH	Approve or reject leave
/Sims/leave-status/	GET	Leave status summary
/Sims/attendance-claims/	GET / POST	List or create correction claims
/Sims/attendance-claims/my-claims/	GET	Own claims
/Sims/attendance-claims/pending-approval/	GET	Claims pending approval
/Sims/attendance-claims/{pk}/approve/	POST	Approve claim
/Sims/attendance-claims/{pk}/reject/	POST	Reject claim with reason

22.5 Tasks & Projects

Endpoint	Method	Description
/Sims/tasks/	GET / POST	List or create tasks
/Sims/tasks/{pk}/	GET / PATCH / DELETE	CRUD on task
/Sims/tasks/due-today/	GET	Tasks due today for current user
/Sims/tasks/{pk}/update-status-auto-next/	PATCH	Move task to next status
/Sims/tasks/assigned-history/	GET	Task assignment history
/Sims/tasks/monthly-count/	GET	Monthly task count for chart
/Sims/tasks/weekly-performance/	GET	Weekly task completion performance
/Sims/team-lead/projects/	GET	Projects for current team lead
/Sims/projects/	GET / POST	List or create projects
/Sims/projects/{pk}/	GET / PATCH / DELETE	CRUD on project
/Sims/projects/dashboard/	GET	Project dashboard stats
/Sims/projects/{pk}/assign-team/	POST	Assign team to project
/Sims/projects/{pk}/assign-team-lead/	POST	Assign team lead
/Sims/teams/	GET / POST	List or create teams
/Sims/teams/{pk}/	GET / PATCH / DELETE	CRUD on team
/Sims/teams/{teamId}/interns/	GET	Interns in team
/Sims/teams/{teamId}/assign-interns/	POST	Assign interns to team
/Sims/teams/{teamId}/remove-intern/	POST	Remove intern from team
/Sims/teams/available-interns/	GET	Interns not yet in a team
/Sims/team-leads/	GET	All team leads
/Sims/completion-review/	GET / POST	Task completion review queue

22.6 Assets, Payments, Documents, Feedback

Endpoint	Method	Description
/Sims/assert-stock/	GET / POST	List or create assets
/Sims/assert-stock/{pk}/	GET / PATCH / DELETE	CRUD on asset
/Sims/assert-stock-count/	GET	Asset count by type/status
/Sims/asset-trend/	GET	Monthly asset trend
/Sims/available-assets/	GET	Available asset count
/Sims/asset-logs/	GET	Asset activity logs
/Sims/asserthistory/	GET	Full asset history
/Sims/asset-lookup/{code}/	GET	Asset lookup by code
/Sims/assert-issue/	GET / POST	Asset issue reports
/Sims/assert-issue/{pk}/replacement-assets/	GET	Available replacements
/Sims/assert-issue/{pk}/assign-new-asset/	POST	Assign replacement
/Sims/fees/	GET / POST	Payment records
/Sims/fees/{empld}/	GET	Intern payment record
/Sims/fees/{pk}/	PATCH	Update payment
/Sims/fee-structure/	GET / POST	Fee structures
/Sims/documents/	GET / POST	Document records
/Sims/documents/{pk}/	GET / PATCH / DELETE	CRUD on document
/Sims/documents/emp/{empld}/	GET	Documents for intern
/Sims/documents/{pk}/download/	GET	Download document as blob
/Sims/approve-document/{pk}/	POST	Approve document
/Sims/reject-document/{pk}/	POST	Reject with comment
/Sims/intern-approval-dashboard/	GET	Approval status per intern

Endpoint	Method	Description
/Sims/feedback/	GET / POST	Feedback records
/Sims/feedback/{id}/	GET / PATCH / DELETE	CRUD on feedback

22.7 Certificates, Dashboard, Notifications, Logs

Endpoint	Method	Description
/Sims/generate-completion-certificate/	POST	Generate completion certificate
/Sims/generate-offer-letter/	POST	Generate offer letter
/Sims/generate-task-certificate/	POST	Generate task certificate
/Sims/generate-attendance-certificate/	POST	Generate attendance certificate
/Sims/generate-partial-certificate/	POST	Generate partial completion certificate
/Sims/dashboard/	GET	General dashboard data
/Sims/performance/	GET	Performance overview
/Sims/api/intern/monthly-performance/	GET	Monthly performance trend
/Sims/admin/dashboard-summary/	GET	Admin dashboard aggregated summary
/Sims/notifications/	GET	User notifications
/Sims/notifications/{id}/mark_read/	PATCH	Mark notification as read
/Sims/notifications/mark_all_read/	POST	Mark all as read
/Sims/logs/	GET	All activity logs
/Sims/logs/{empld}/	GET	Activity logs for intern
/Sims/onboarding/list/	GET	Pending onboarding submissions
/Sims/onboarding/enable/{responseld}/	POST	Approve and activate intern
/Sims/onboarding/send_credentials/{internId}/	POST	Send login credentials

22.8 Forms App (/api/ prefix)

Endpoint	Method	Description
/api/forms/	GET / POST	List or create forms

Endpoint	Method	Description
/api/forms/{id}/	GET / PATCH / DELETE	CRUD on form
/api/forms/{id}/upload_banner/	POST	Upload form banner
/api/forms/{id}/analytics/	GET	Form response analytics
/api/forms/{id}/responses/	GET	All responses for form
/api/public/feedback/{formId}/	GET / POST	Public form submission
/api/responses/{responseId}/consult/	PATCH	Toggle consult flag
/api/responses/{responseId}/comments/	GET / POST	Response comments

23. Complete End-to-End Execution Flow

The complete lifecycle of a SIMS deployment from first install to intern exit, in chronological operational order.

Phase 1 — System Setup (Super Admin)

- 89. Super Admin logs in and creates an Entity (company/academy name).
- 90. Creates a Manager account and assigns them to the entity.
- 91. Activates required modules (Attendance, Tasks, Payments, AI Features).

Phase 2 — Organization Configuration (Manager)

- 92. Creates Branches, Departments, and Domains.
- 93. Configures working hours, shifts, leave policy, payment cycles, and AI feature toggles in Settings.
- 94. Creates Lead and Mentor accounts via /admin/register.
- 95. Creates Teams and assigns Mentors as team leads.

Phase 3 — Intern Onboarding

- 96. Intern submits onboarding form at /InternOnboarding.
- 97. AI Onboarding Readiness Check automatically evaluates submission completeness and resume domain fit.
- 98. Manager reviews (with AI readiness score) and approves the account.
- 99. Manager sends login credentials to intern.
- 100. Intern logs in, uploads required documents. AI evaluates uploaded resume.
- 101. Staff reviews and approves documents.
- 102. Manager assigns intern to Domain and Team. Issues assets.
- 103. Intern status → Active.

Phase 4 — Learning Phase

- 104. Mentor assigns learning tasks. AI generates a personalized learning path based on resume skill analysis.
- 105. Intern works through learning tasks. AI generates practice quizzes and explains concepts on demand.
- 106. Intern uses AI Mock Interview for technical interview preparation.
- 107. Intern uses AI Resume Builder to create a professional resume from SIMS data.
- 108. Mentor reviews learning progress. AI flags if intern is behind pace.
- 109. Learning phase marked complete → intern transitions to Active project status.

Phase 5 — Active Internship (Daily Operations)

- 110. Intern checks in via attendance widget. AI chatbot available for questions and navigation.
- 111. Mentor/Lead assigns project tasks. AI suggests task descriptions, subtask breakdowns, and SLA estimates.
- 112. Intern works on tasks, updating status. AI explains any unclear task requirements.
- 113. Intern checks out at end of day. Total hours calculated.
- 114. Leave application → Mentor approval via attendance module.
- 115. AI Attendance Anomaly Detection runs daily — flags unusual patterns to Mentor.

Phase 6 — Ongoing Monitoring (Staff)

- 116. Mentor reviews Intern Management Dashboard with AI performance badges and risk flags.
- 117. Lead reviews Task Dashboard with AI project health indicators.
- 118. SLA warnings appear on overdue tasks. Escalation chain triggers automatically.
- 119. Mentor opens Performance Feedback — AI drafts feedback from metrics. Mentor reviews and submits.
- 120. Manager reviews AI report: 'Which interns are at risk this month?' via natural language report generator.
- 121. Manager processes payroll via Payroll Dashboard.

Phase 7 — Performance Evaluation

- 122. Lead reviews AI Performance Analysis report for each intern.
- 123. AI identifies top performers, at-risk interns, and outliers with reasoning.
- 124. Lead performs final evaluation using AI-suggested evaluation summary as a starting draft.
- 125. Lead marks outcome: Selected / Not Selected.
- 126. Eligible interns promoted: Free → Paid → Stipend → Employee.

Phase 8 — Exit & Certificate Generation

- 127. Internship end date reached or Manager marks Complete.
- 128. Intern returns all assets. Condition verified. Asset record closed.
- 129. AI generates Exit Summary & Career Readiness Report for the intern.
- 130. Manager generates Completion Certificate (standard + AI-enhanced version with career readiness report).
- 131. Intern status → Completed. Account archived (soft delete).
- 132. All data retained. Final reports available in analytics.

24. Complete Feature Checklist

Definitive checklist for the complete system including all AI features. Use as acceptance criteria for project completion.

24.1 Identity & Access

- Multi-entity architecture with Super Admin
- Six-level role hierarchy: Super Admin, Manager, Lead/SME, Mentor, Intern (Staff View), Intern (Self)
- Token-based auth with global 401 interceptor and auto-logout
- OTP-based password reset flow
- Role-based route protection (ProtectedRoute + ProtectedDashboardRoute)
- Server-side data visibility scoping per role
- Entity-level module activation/deactivation (including AI features)
- Per-role permission policy customization by Manager
- Role delegation with time expiry and attribution logging
- MFA (TOTP/OTP) for Manager and above
- Session management: active session list, force logout
- Password complexity enforcement and expiry reminders

24.2 Organization & Users

- Entity, Branch, Department, Domain hierarchy with full CRUD
- Staff account creation and management
- Intern multi-step onboarding form with all required fields
- Onboarding approval workflow with credential dispatch
- Profile update request and approval flow
- CSV bulk import for interns, tasks, and assets with validation and error log
- Soft delete and recovery for all user types
- Team creation and intern assignment management

24.3 Intern Lifecycle

- Lifecycle states: yettojoin, active, onleave, completed, discontinued
- Enforced state transitions with history tracking
- Promotion pathway: Free → Paid → Stipend → Employee → Mentor/Lead
- Promotion approval workflow with date and basis recording
- Archive function for completed/discontinued interns
- Activity timeline per intern: chronological event feed of all actions
- Notes/comments on intern profiles: internal vs intern-visible flag

24.4 Attendance & Leave

- Check-in / check-out with session timer and break tracking
- Daily attendance log with total hours and effective hours calculation
- Leave application and multi-level approval workflow
- Leave balance tracking and quota enforcement
- Leave deduction integrated into attendance % calculation
- Attendance correction claim requests with approval
- Shift definitions per entity with late-mark and half-day rules
- Intern hours calculator

24.5 Task & Project Management

- Task CRUD with type, priority, assignee, project, due date, subtasks
- Task status flow with completion review queue
- Task dependency (blocked-by) with transition guard
- SLA deadline fields with visual warning and breach indicators
- Auto-escalation engine: overdue → intern → Mentor → Lead → Manager
- Task versioning: full edit history with diff view
- Task-level timer for effective hours tracking
- Bulk project task creation via grid interface
- Project CRUD with team and team lead assignment
- Gantt chart view for project planning
- Team management with intern assignment/removal

24.6 Learning Phase

- Dedicated learning task type isolated from project tasks
- Sequential learning path assignment with dependency ordering
- Intern-facing learning progress view separate from project tasks
- Mentor visibility of per-intern learning completion %
- AI-generated personalized learning path from skill gap analysis
- AI practice quiz generation from learning task content
- AI concept explainer within learning tasks

24.7 Payments

- Three payment types: Free, Paid, Stipend (set at onboarding)
- Payment CRUD with status: Paid, Pending, Overdue
- Cash payment approval workflow
- Payment history versioning
- Payment cycle configuration per entity

- Intern read-only payment view

24.8 Asset Management

- Asset CRUD with code, type, serial number, condition, status
- Asset assignment with issue date and expected return date
- Issue reporting with replacement workflow
- Return with condition verification
- Full audit history per asset
- Soft delete and restore

24.9 Document Management

- Document upload with type classification
- Validation pipeline: Pending → Approved / Rejected with comments
- Re-upload flow for rejected documents with version history
- Structured file storage hierarchy with type/size validation
- Document download as blob
- AI resume evaluation integrated into upload flow

24.10 Feedback & Performance

- Mentor periodic feedback with qualitative notes and numeric scores
- Lead final evaluation with Selected / Not Selected outcome
- Student-to-staff feedback
- Public feedback forms with shareable link
- Forms builder with 8 question types
- Auto-calculated metrics: attendance%, completion rate, effective hours ratio, quality rating
- Feedback versioning
- Intern visibility of final evaluation (configurable)

24.11 AI Features

- AI Chatbot Assistant — LLM-powered, voice input/output, context-aware, multi-turn
- AI Performance Analysis — narrative report, scoring, risk flags, comparative analytics
- AI Learning Path Generator — skill gap analysis, personalized path, domain readiness score
- AI Practice Quiz Generator — MCQ/short-answer from learning task content
- AI Concept Explainer — explain any term in a task description
- AI Resume Evaluator — ATS score, domain fit, section scores, improvement suggestions
- AI Resume Builder — generates complete resume from SIMS data, exportable as PDF
- AI Mock Interview — domain-specific, behavioral, project-based, company prep interview modes

- AI Interview Report — per-question scores, strengths, improvement areas, action plan
- AI Smart Task Suggestions — description generator, task decomposer, SLA estimator, workload balancer
- AI Attendance Anomaly Detection — pattern detection, dropout risk scoring
- AI Feedback Writing Assistant — draft generator, tone checker, consistency checker
- AI Exit Summary & Career Readiness Report — exportable PDF for job applications
- AI Smart Search — natural language queries across all entities
- AI Onboarding Readiness Check — submission completeness and domain fit scoring
- AI Natural Language Report Generator — any report from a plain English request
- AI Notification Intelligence — smart digest, priority classification, noise reduction

24.12 Certificates

- Completion, Offer Letter, Task, Attendance, Partial Completion certificates
- AI-Enhanced Completion Certificate with Career Readiness Report

24.13 Notifications

- In-app notifications with unread count badge
- Email notifications for key events
- AI smart digest, priority classification, contextual summaries
- Per-user notification preferences

24.14 Reporting & Analytics

- Role-based dashboards: Admin, Task, Attendance, Asset, Payroll, Intern Self
- CSV/Excel export from all major list views
- AI-generated PDF reports from natural language requests
- Activity/audit log viewer
- AI Performance Scoring and comparative analytics
- AI risk flags: at-risk interns, overloaded interns, disengaged interns

24.15 System-Wide

- Soft delete and recovery for all major entities
- Versioning and edit history for tasks, payments, roles, documents
- Immutable audit log with AI interaction audit trail
- Unified calendar view: task deadlines, leaves, milestones
- iCal feed for Google Calendar / Outlook sync
- Global AI smart search across all entities
- Advanced filter save/bookmark on all list views

- UTC timestamp storage with local timezone display
- REST API with documented endpoints
- Webhooks for key events
- Dark mode persisted to user profile
- Responsive design for all core intern flows on mobile
- Notes/comments on tasks and intern profiles

25. Complete Page & Component Map

26.1 Public / Auth Pages

Page / Component	Route	Purpose
InternHomePage	/	Public landing page with hero slider, features, about, contact, program banner.
LoginPage	/loginpage	Login form. Routes to appropriate dashboard on success.
Recovery	/Recovery	Step 1 of password reset: enter email, receive OTP.
Reset	/Reset	Step 2 of password reset: enter OTP and new password.
AboutUs1	/AboutUs1	Public about page.
ContactUs1	/ContactUs1	Public contact page.
PublicFeedbackForm	/feedback/:formId	Public form response submission (no auth required).

26.2 Admin Dashboard Pages (under /admin/*)

Component	Route	Purpose
AdminDashboardContent	/admin/dashboard	Main summary dashboard for Manager/Super Admin.
StaffList	/admin/user	All staff management: view, edit, deactivate.
InternLists	/admin/interns	All intern management with CSV export and bulk actions.
PaymentList	/admin/payments	Payment records management.
DepartmentManagement	/admin/departments	Department and domain hierarchy CRUD.
EntityManagement	/admin/entities	Entity, branch, entity-department hierarchy CRUD.
PerformanceFeedbackPage	/admin/feedback	All feedback submissions.
TeamManagement	/admin/teams	Team creation and composition.
RegisterPage	/admin/register	Staff registration form.
AdminProfile	/admin/profile	Admin/Manager profile management.

26.3 Intern Management Dashboard Pages (under /intern/*)

Component	Active View	Purpose
InternDashboard (dashboard)	/intern/dashboard	Summary cards and intern overview.
InternManagementLists	/intern/users	Intern table with status, domain, actions.
DocumentView (staff)	/intern/documents	Document upload and approval per intern.
PerformanceFeedbackList	/intern/feedbacks	Feedback list for managed interns with AI draft button.
Forms	/intern/forms	Form builder and response viewer.
InternApprovalDashboard	/intern/approval-dashboard	Pending document approvals.
CompletionList	/intern/approvals (completionList)	Task completion review queue.

26.4 Task Dashboard Pages (under /task/*)

View	Active View ID	Purpose
Tasks Dashboard	tasks	Task overview: counts, charts, due-today list, AI workload insights.
Task List	task-list	Full task table with filters, bulk actions, AI suggestion button.
Projects	projects	Project cards with status, team, task count. Full CRUD.
Project Status	project-status	Per-project detail with task breakdown, team, progress bar, milestones.
Teams	teams	Team composition and intern assignment.
Departments	departments	Department and domain hierarchy view.
Individual Task	individual-task	Single task deep-dive with subtasks, comments, file attachments.
User List	user-list	User list in task context.

26.5 Attendance, Payroll & Asset Dashboard Pages

Dashboard	Views	Route
AttendanceDashboard	dashboard, table (Daily), attendance-log, leave, attendance-claims, profile	/attendance/*
PayrollDashboard	dashboard, payments (Payment Management), profile	/payroll/*
AssetDashboard	dashboard, assets (Asset Management), intern-status, asset-report, profile	/asset/*

26.6 Intern Self-Service Pages (under /intern-user/*)

Page Component	Route	Purpose
Dash.js (Dashboard)	/intern-user/dashboard	Home: check-in/out, tasks, announcements, performance summary, AI chatbot.
AttendanceManagement	/intern-user/attendance	Check in/out, breaks, attendance log.
DocumentView	/intern-user/documents	Upload documents, view approval status. AI resume evaluation.
PaymentStatusPage	/intern-user/payment-status	Read-only payment record view.
AssetReport	/intern-user/asset-report	View assigned assets, submit issue reports.
PerformancePage	/intern-user/performance-feedback	AI performance analysis, metrics, feedback charts.
LeaveManagement	/intern-user/leave	Apply for leave, view history and balance.
Tasks (read-only)	/intern-user/tasks	View and update own task list.
InternHoursCalculator	/intern-user/hours-calculator	Calculate total effective working hours for any date range.
StudentStaffFeedback	/intern-user/staff-feedback	Submit feedback about Mentor/Lead.
TeamsManagement	/intern-user/teams	View team composition and Mentor.
InternProfile	/intern-user/profile	Own profile view and edit.
MockInterviewPage (NEW)	/intern-user/mock-interview	AI mock interview sessions and history.
ResumeBuilderPage (NEW)	/intern-user/resume-builder	AI resume generator, evaluator, and PDF export.

Page Component	Route	Purpose
LearningPage (NEW)	/intern-user/learning	AI learning path, quizzes, and concept explainer.

26.7 New Staff/Admin AI Pages

Page	Route	Purpose
InternHomePage	/	Public landing page
LoginPage	/loginpage	Login form
Recovery	/Recovery	Password reset step 1
Reset	/Reset	Password reset step 2
PublicFeedbackForm	/feedback/:formId	Public form submission
AdminDashboard (shell)	/admin/*	Admin shell: routes to sub-components
AdminDashboardContent	/admin/dashboard	Admin summary dashboard
StaffList	/admin/user	Staff management
InternLists	/admin/interns	Intern management with export
PaymentList	/admin/payments	Payment records
DepartmentManagement	/admin/departments	Department/domain hierarchy
EntityManagement	/admin/entities	Entity/branch hierarchy
PerformanceFeedbackPage	/admin/feedback	All feedback
TeamManagement	/admin/teams	Team management
RegisterPage	/admin/register	Staff registration
InternDashboard	/intern/*	Intern management shell
Tasks	/task/*	Task management shell
AttendanceDashboard	/attendance/*	Attendance management shell
PayrollDashboard	/payroll/*	Payroll management shell
AssetDashboard	/asset/*	Asset management shell
Dash.js	/intern-user/dashboard	Intern home dashboard
AttendanceManagement	/intern-user/attendance	Intern check-in/out
DocumentView	/intern-user/documents	Intern document upload
PaymentStatusPage	/intern-user/payment-status	Intern payment view
AssetReport	/intern-user/asset-report	Intern asset view

Page	Route	Purpose
PerformancePage	/intern-user/performance-feedback	Intern performance view
LeaveManagement	/intern-user/leave	Intern leave management
InternHoursCalculator	/intern-user/hours-calculator	Hours calculator
StudentStaffFeedback	/intern-user/staff-feedback	Staff feedback submission
MockInterviewPage (NEW)	/intern-user/mock-interview	AI mock interview sessions
ResumeBuilderPage (NEW)	/intern-user/resume-builder	AI resume builder and evaluator
LearningPage (NEW)	/intern-user/learning	AI learning path and quizzes
AIReportsPage (NEW)	/admin/ai-reports	Natural language report generator
AllInsightsPage (NEW)	/intern/ai-insights	AI performance analysis for all interns
AuditLogPage (NEW)	/admin/audit-log	Immutable audit log viewer
DelegationPage (NEW)	/admin/delegation	Role delegation management
CalendarPage (NEW)	/calendar	Unified calendar view
ImportPage (NEW)	/admin/import	CSV bulk import wizard
WebhooksPage (NEW)	/admin/webhooks	Webhook configuration

27. Key Data Models

Reference of the primary data models inferred from the API structure and frontend field usage.

27.1 User / Intern Record

Field	Type	Notes
emp_id	String (PK)	Employee/Intern unique identifier.
username / email	String	Login credential.
role	Enum	superadmin / manager / lead / mentor / intern.
user_status	Enum	yettojoin / inprogress / onleave / completed / discontinued.
department	FK → Department	Assigned department.
domain	FK → Domain	Assigned domain/specialization.
scheme	Enum	Free / Paid / Stipend.

Field	Type	Notes
shift_timing	String	Assigned shift (e.g., Morning, Evening, Standard).
start_date / end_date	Date	Internship duration.
photo	File	Profile photo.
is_deleted	Boolean	Soft delete flag.

27.2 Task

Field	Type	Notes
id	Integer (PK)	Task ID.
title / description	String / Text	Task name and detailed requirements.
status	Enum	To Do / In Progress / Completed.
priority	Enum	Low / Medium / High.
task_type	Enum	Learning / Project / Custom.
assigned_to	FK → User	Responsible intern or staff.
project	FK → Project	Parent project.
due_date / committedDate	Date	Task deadline for SLA calculation.
progress	Integer (0–100)	Progress percentage.
blocked_by	FK → Task (self)	Dependency — prerequisite task.
sla_hours	Integer	SLA time window in hours.
subtasks	Array	Child tasks with own status.

27.3 Attendance Record

Field	Type	Notes
emp_id	FK → User	Intern the record belongs to.
date	Date	Attendance date.
check_in	DateTime	Check-in timestamp.
check_out	DateTime	Check-out timestamp.
total_hours	Float	Calculated: check_out – check_in – break_time.
break_start / break_end	DateTime	Break window timestamps.

Field	Type	Notes
status	Enum	Present / Absent / On Leave.

27.4 Asset

Field	Type	Notes
asset_code	String (unique)	Lookup key for asset.
asset_type	String	Laptop, ID Card, Accessories, etc.
serial_number	String	Physical serial number.
condition	Enum	Good / Damaged / Lost.
status	Enum	Available / Assigned / Damaged / Lost.
assigned_to	FK → User	Currently assigned intern.
issue_date	Date	Date asset was issued.
expected_return_date	Date	Expected return date.
is_deleted	Boolean	Soft delete flag.

28. New Components Required

- AIChatWidget — floating chatbot widget with voice input/output (upgrade existing rule-based chatbot)
- AIPerformanceReport — full-page AI narrative performance analysis with export
- AIScoreBadge — compact AI performance score chip shown on intern cards and lists
- AIRiskFlag — visual risk indicator (at-risk, overloaded, disengaged) with tooltip explanation
- MockInterviewPanel — full interview session UI with question display, answer input, real-time score chips
- InterviewReportCard — structured report display with per-question breakdown and charts
- ResumeEditor — section-by-section resume editor with AI improve buttons per section
- ResumeSectionSuggestion — side panel showing AI suggestions for a selected resume section
- ResumePreview — real-time styled preview of the generated resume
- LearningPathTimeline — visual sequential learning path with locked/unlocked state per task
- AIQuizCard — MCQ/short-answer quiz interface with immediate answer feedback
- SkillGapChart — radar/bar chart showing current vs required skills for the domain
- TaskDependencySelector — multi-select tasks as blocked-by prerequisites
- SLABadge — countdown chip showing time remaining / breach status on task cards
- EscalationTimeline — shows escalation chain status for overdue tasks
- VersionHistoryDrawer — side panel with field-level diff history for tasks/payments/documents
- ActivityTimeline — chronological event feed per intern
- CommentThread — threaded comments with internal/visible flag on tasks and profiles
- BulkImportWizard — step-by-step CSV import with column mapping, preview, error report
- CalendarWidget — unified calendar with category filters (tasks, leaves, milestones)
- PromotionModal — promotion approval workflow with basis recording
- GlobalSearchBar — AI-powered search overlay with scoped results
- DelegationModal — create delegation: user, permissions subset, date range
- ShiftConfigEditor — entity-level shift definition with late-mark rules
- EntityFeatureFlagPanel — toggle module and AI feature activation per entity
- NaturalLanguageReportInput — text input that triggers AI report generation

26. UI/UX Guidelines & Design System

29.1 Technology

Complete frontend technology stack:

- Primary UI: Material UI (MUI) v6 with custom theme — consistent typography, spacing, elevation across all dashboards
- Utility styling: Tailwind CSS v4 for layout, spacing, and responsive utilities
- Animation: Framer Motion for page transitions and micro-interactions
- Drag & Drop: @dnd-kit/core and @dnd-kit/sortable for Kanban task boards
- Date/Time: MUI X Date Pickers with dayjs adapter — all date display uses dayjs for consistent formatting
- Charts: MUI X Charts, Chart.js (via react-chartjs-2), Recharts — each used consistently per chart type
- Icons: MUI Icons Material + Lucide React + Heroicons
- Data Tables: MUI DataGrid (x-data-grid v7) for all large tabular data with built-in pagination, sorting, column resizing
- Circular progress: react-circular-progressbar for performance metric display
- CSV Export: react-csv for intern list and report exports
- Status chips: MUI Chip with color-coded states throughout all list views
- Priority badges: Low (green), Medium (orange), High (red)

29.2 Layout Pattern

- All dashboards: fixed header (64px) + fixed left sidebar (240px) + scrollable main content area
- Mobile: sidebar collapses to a hamburger menu. Core intern flows (check-in, tasks, leave) are fully usable on mobile
- Admin dashboard uses a simplified sidebar with only 5 navigation items for clean management UX
- AI features surface contextually — not as a separate menu but as enhancement buttons within relevant pages

29.3 Dark Mode

- Dark mode is toggled via ThemeContext (ThemeToggle.js component) stored in Settings
- Dark mode preference is persisted to the user profile (not just local state) — survives logout/re-login
- Dark background color: #0f0f1a. Light background: bg-gray-50

29.4 Feedback Patterns

- Toast notifications: react-toastify (top-right, 5 second auto-close) for operation success/error

- Inline snackbars (MUI Snackbar) for context-specific alerts within dashboards
- Loading spinners (MUI CircularProgress) on all data-fetching operations
- Empty state illustrations for lists with no data
- Confirmation dialogs (MUI Dialog) for destructive actions (delete, deactivate, reject)
- Streaming text display for AI responses — animated typewriter effect via character buffering

29.5 Data Display

- MUI DataGrid (x-data-grid v7) for all large tabular data with built-in pagination, sorting, and column resizing
- MUI X Charts, Chart.js (via react-chartjs-2), and Recharts for all visualizations — used consistently per chart type
- Circular progress bars (react-circular-progressbar) for performance metric display
- Status chips (MUI Chip) with color-coded states throughout all list views
- Priority badges: Low (green), Medium (orange), High (red)

29.6 AI Feature UX Principles

- Primary UI: Material UI (MUI) v6 — consistent typography, spacing, elevation across all dashboards
- Utility styling: Tailwind CSS v4 for layout, spacing, and responsive utilities
- Animation: Framer Motion for page transitions and micro-interactions
- Drag & Drop: @dnd-kit/core and @dnd-kit/sortable for Kanban task boards
- Date/Time: MUI X Date Pickers with dayjs adapter — all dates display via dayjs for consistency
- Charts: MUI X Charts, Chart.js (react-chartjs-2), Recharts — use consistently per chart type
- Icons: MUI Icons Material + Lucide React + Heroicons
- PDF Export: react-signature-canvas, exceljs, file-saver, xlsx for export capabilities

29.7 Layout Principles

- All dashboards: fixed header (64px) + fixed left sidebar (240px) + scrollable main content area
- Mobile: sidebar collapses to hamburger. Core intern flows (check-in, tasks, leave) fully mobile-usable
- Admin dashboard uses simplified 5-item sidebar for clean management UX
- AI features surface contextually — not as a separate menu but as enhancement buttons within relevant pages

29.8 AI Feature UX Principles

- AI suggestions are never auto-applied — always presented as a draft or suggestion the user can accept/edit/ignore
- AI responses stream where possible — avoid blank wait screen by showing tokens as they arrive

- AI actions are clearly labeled with a  icon or 'AI' badge so users know when AI is involved
- Every AI output has a 'Regenerate' and 'Feedback' option so users can improve results
- AI performance score is shown alongside (never instead of) the raw metric values
- AI chatbot is a floating widget — non-intrusive, dismissed with one click, never auto-opens

29.9 Feedback Patterns (Detail)

- Toast notifications (react-toastify, top-right, 5s) for operation success/error
- Inline MUI SnackBar for context-specific alerts within dashboards
- MUI CircularProgress on all data-fetching operations
- Empty state illustrations for lists with no data
- MUI Dialog for confirmation on destructive actions (delete, deactivate, reject)
- Streaming text display for AI responses — animated typewriter effect via character buffering

End of SIMS Complete Project Specification with AI Features

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